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FIXED INCOME SECURITIES – PART 2



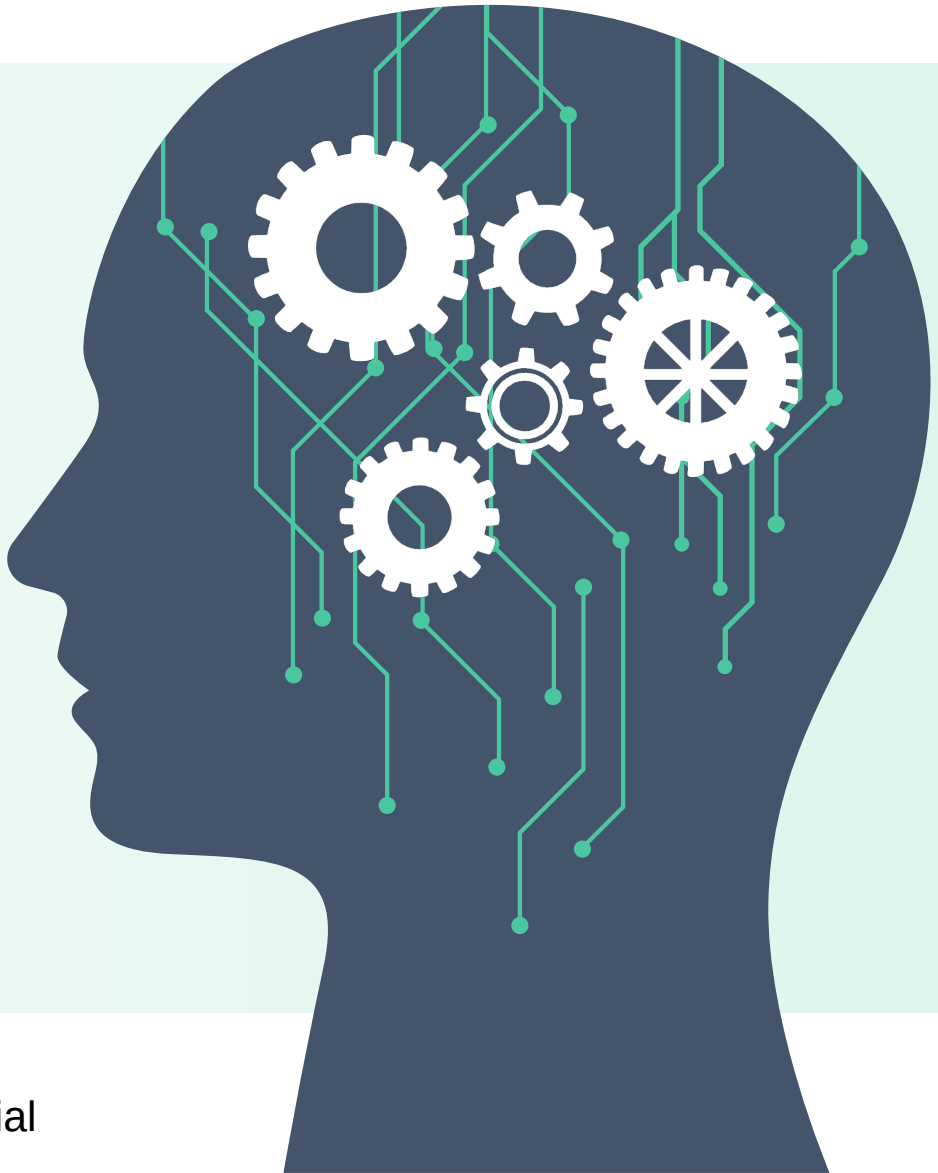
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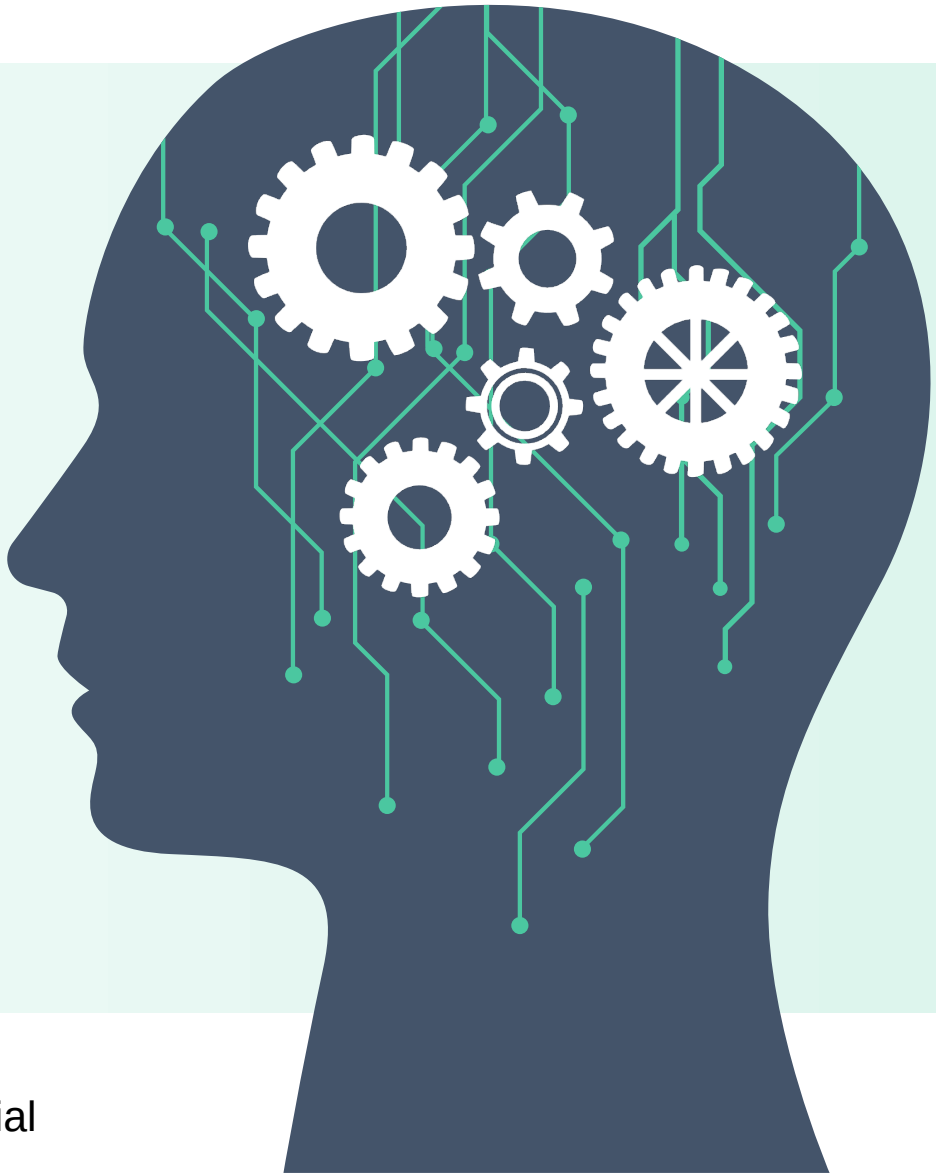
YOU ARE GOING TO LEARN…

- Primary Market - Fixed Income Securities Issues
- Secondary Market - Fixed Income Securities Trading
- Pricing of Bonds, Clean Price vs Dirty Price
- Relationship between Market Interest Rates & Bond Prices



YOU ARE GOING TO LEARN…

- Bond Yields (Returns) : Current Yield & YTM
- Securitization Process
- Securitized Products





Primary Issuance of Bonds

PRIMARY ISSUANCE OF BONDS

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Bonds are offered to the public through various methods and channels in the primary market. The process can vary depending on the type of bond, the issuer, and market conditions.

Here are some common ways in which bonds are offered to the public:

1. **Public Auction:** In some cases, especially for government bonds, bonds are offered to the public through a competitive bidding process. Investors submit bids specifying the quantity of bonds they want to purchase and the yield (interest rate) they are willing to accept. The bonds are then allocated to the highest bidders until the entire issue is sold. This method is common for U.S. Treasury securities.

2. **Private Placement:** Some bonds are offered directly to a select group of institutional investors or accredited individuals in a private placement. This method is common for corporate bonds, especially those with specific terms or a smaller issuance size. Private placements may have fewer regulatory requirements and can be tailored to the needs of the issuer and investors.

PRIMARY ISSUANCE OF BONDS

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3 **Retail Offerings:** Bonds can be offered directly to retail investors through banks, financial institutions, and brokerage firms. Retail investors can purchase bonds in smaller denominations, making them accessible to a broader range of individual investors. Retail offerings may include government savings bonds and corporate bonds.

4 **Underwritten Offerings:** Many bond issuances involve underwriters, typically investment banks or financial institutions, which purchase the bonds from the issuer and then resell them to investors. Underwriters play a key role in marketing the bonds, determining pricing, and managing the issuance process. They may offer the bonds to institutional investors, retail investors, or a combination of both.

5 **Syndicated Offerings:** Large bond issuances, especially those by corporations or governments, often involve a syndicate of underwriters working together to market and distribute the bonds. Each underwriter in the syndicate has a specific role in selling the bonds to investors. Syndicated offerings are common for high-value corporate bonds and government bonds.

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6 **Online Platforms:** With the growth of online investing platforms and electronic trading, many bonds are offered to the public through digital channels. Investors can access bond offerings and make purchases through online brokerage accounts, making it easier for individuals to participate in bond markets.

7 **Direct Purchase Programs:** Some issuers, especially municipalities and corporations, offer direct purchase programs that allow investors to buy bonds directly from the issuer without going through an intermediary. These programs are designed to make it easier for individual investors to buy bonds.

8 **Book-Building Process:** In many bond offerings, especially those involving corporate bonds, the underwriters use a book-building process. During this process, they gauge investor demand and set the final bond pricing based on the orders submitted by investors. This helps determine the bond's yield and the allocation of bonds to investors.

Bonds can be issued at a discount, at par, or at a premium, and the choice of issuance price depends on a combination of factors, including market conditions, prevailing interest rates, and the issuer's objectives.

PRIMARY ISSUANCE OF BONDS

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1. Issued at a Discount:

- **Reason:** Bonds are often issued at a discount when market interest rates are higher than the coupon rate of the bond being issued. In other words, the bond's stated interest rate (coupon rate) is lower than the prevailing market interest rates.
- **Effect:** When a bond is issued at a discount, investors pay less than the bond's face value (par value) to purchase it. The difference between the face value and the issuance price represents the discount.
- **Impact:** Discount effectively serves as additional capital gain to compensate for the lower coupon rate if the bonds are redeemed at Par/Face Value. Issuing bonds at a discount allows the issuer to reduce the cost of borrowing.

2. Issued at Par:

- **Reason:** Bonds are typically issued at par when the coupon rate of the bond matches the prevailing market interest rate. In such cases, the bond's coupon rate is considered competitive, and investors are willing to buy the bonds at face value.
- **Effect:** Bonds issued at par are sold to investors at their face value, and the interest payments (coupon payments) are equal to the coupon rate multiplied by the face value.
- **Impact:** The issuer receives the full face value of the bond, and investors earn a predictable stream of interest income without any capital gain or loss upon maturity.

PRIMARY ISSUANCE OF BONDS

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3. Issued at a Premium:

- **Reason:** Bonds may be issued at a premium when market interest rates are lower than the coupon rate of the bond. In this case, investors are willing to pay more than the bond's face value to secure the higher coupon payments.
- **Effect:** Bonds issued at a premium are sold to investors at a price higher than their face value. The amount by which the price exceeds the face value represents the premium.
- **Impact:** Premium effectively reduces the overall yield to maturity (YTM) for investors if the Bonds are redeem at Par. Issuing bonds at a premium allows the issuer to offer a higher coupon rate while keeping the effective cost of borrowing lower.

In summary, the issuance price of a bond is influenced by the relationship between the bond's coupon rate and prevailing market interest rates. Bonds are issued at a discount when coupon rates are lower than market rates, at par when coupon rates are competitive, and at a premium when coupon rates are higher than market rates. The choice of issuance price helps issuers attract investors and manage their borrowing costs. It's important to note that the issuance price, along with the bond's coupon rate and term to maturity, determines the overall yield to investors.



Bonds - Secondary Market

THE SECONDARY BOND MARKET

The secondary market for bonds, often referred to as the bonds trading market or the fixed-income securities market, is where already-issued bonds are bought and sold among investors after their initial issuance in the primary market. In the secondary market, bonds are traded on various exchanges and over-the-counter (OTC) platforms.

Key Participants:

1. Investors

This includes individual investors, institutional investors (such as mutual funds, pension funds, and insurance companies), and traders who buy and sell bonds for profit.

2. Brokers and Dealers

Brokerage firms and bond dealers act as intermediaries between buyers and sellers, facilitating bond transactions. They often maintain an inventory of bonds for sale.

3. Exchanges and OTC Platforms

Bonds are traded on established exchanges like the New York Stock Exchange (NYSE) / NSE or through OTC platforms like the bond desks of major financial institutions.

THE SECONDARY BOND MARKET

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Characteristics and Operations:

1. Liquidity

The secondary bond market provides liquidity to bondholders. Investors can sell their bonds and exit their investments before the bonds mature. This liquidity is one of the primary reasons why bonds are attractive investments.

2. Price Determination

Bond prices in the secondary market are influenced by various factors, including changes in interest rates, credit quality, market conditions, and supply and demand dynamics. When interest rates rise, bond prices tend to fall, and vice versa.

3. Yield

The yield on a bond, which represents the return an investor earns, is also determined in the secondary market. As bond prices change, yields adjust accordingly. If a bond's price falls, its yield rises, and vice versa.

4. Speculation

Bonds can be traded actively in the secondary market, and investors can engage in buying and selling bonds frequently. This trading activity allows investors to adjust their portfolios and take advantage of market conditions.

THE SECONDARY BOND MARKET

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Characteristics and Operations:

5. Market Participants

While institutional investors dominate the secondary bond market, individual investors can also participate through brokers and mutual funds that invest in bonds.

6. Market Segments

The bond market is diverse and includes various segments, such as government bonds, corporate bonds, municipal bonds, and mortgage-backed securities. Each Bond Market Segment has its own characteristics and market participants.

7. Transparency

Bond prices and transaction information are often reported and publicly available, providing transparency to market participants.

THE SECONDARY BOND MARKET

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Trading Mechanisms:

1. Exchange-Traded Bonds

Some bonds are listed on exchanges and trade much like stocks. Investors can place market orders or limit orders to buy or sell these bonds during trading hours.

2. Over-the-Counter (OTC) Market

The majority of bond trading occurs in the OTC market. Bonds are traded directly between buyers and sellers or through bond dealers. These transactions are often negotiated over the phone or electronically.

THE SECONDARY BOND MARKET

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The secondary bond market is crucial for several reasons:

1. **Liquidity:** It provides liquidity to bondholders, allowing them to convert their investments into cash before maturity.
2. **Price Discovery:** Bond prices in the secondary market help determine the market value of existing bonds, influencing their pricing in primary offerings.
3. **Risk Management:** Investors use the secondary market to adjust their bond portfolios in response to changing market conditions and to manage interest rate risk and credit risk.
4. **Diversification:** Bonds as an Asset Class allows a wide range of investors, including individuals, to access and invest in bonds, diversifying their investment portfolios.
5. **Capital Formation:** The secondary market indirectly supports capital formation by providing a liquid exit for investors, making bonds more attractive investment instruments.

DIRTY PRICE & CLEAN PRICE



The price which the bond buyer pays the bond seller which includes the Accrued Interest

Dirty Price = Clean Price + Accrued Interest

Clean Price is the Market Value of the bond at which which the Bond is

Trading i.e buyer is willing to pay to the seller

Excludes accrued interest

CLEAN PRICE / FLAT PRICE & DIRTY PRICE

Clean price and dirty price are terms commonly used in the bond market to describe the price of a bond and its associated accrued interest. They are also known as "**Quoted price / Flat Price**" and "**Invoice Price**," respectively. These terms are essential for understanding the actual cost or value of a bond transaction.



CLEAN PRICE / FLAT PRICE & DIRTY PRICE

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Clean / Flat Price (Quoted Price):	Dirty Price (Invoice Price):
The clean price of a bond refers to the market price of the bond without including any accrued interest.	The dirty price, also known as the invoice price or full price or total cost price, includes both the market price of the bond and the accrued interest.
It represents the price at which the bond is traded in the secondary market.	The dirty price accounts for any interest that has accrued on the bond since the last interest payment date. This accrued interest is typically added to the clean price to calculate the dirty price.
The clean price is quoted as a percentage of the bond's face value (par value) or on a per-bond basis. For example, if a bond's clean price is quoted as 98.50, it means the bond is priced at 98.50% of its face value.	Bond transactions often settle on the basis of the dirty price, so the buyer pays the accrued interest to the seller in addition to the clean price.
The clean price is the price that the buyer pays to the seller for the bond itself, excluding any interest that has accrued since the last interest payment date.	The dirty price allows bond investors to determine the actual cost of buying a bond, taking into account the interest earned by the seller up to the settlement date.

DIRTY PRICE CALCULATION

The formula to calculate the dirty price is as follows:

Dirty Price = Clean Price + Accrued Interest

Where:

Clean Price is the market price of the bond.

Accrued Interest is the interest that has accumulated on the bond from the last interest payment date up to the settlement date.

Investors and traders in the bond market use both clean and dirty prices to assess the value of bonds, especially when buying or selling bonds between interest payment dates. The distinction between the two prices is important because it helps investors understand the total cost or value of a bond transaction, taking into consideration the accrued interest that must be paid by the buyer or received by the seller.

DIRTY PRICE CALCULATION

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Accrued Interest:

- Accrued interest refers to the interest that has accumulated on a bond since the last interest payment date.
- When a bond pays periodic interest, typically semiannually or annually, interest accrues daily between interest payment dates.
- It is the compensation that the bondholder (the lender) earns for lending their money to the issuer (the borrower) for a portion of the interest period.
- Accrued interest is important when bonds are bought or sold between interest payment dates because the buyer must compensate the seller for the interest that has accrued up to the settlement date.

Day Count Convention:

- Day count convention, also known as the day-count basis or day-count fraction, is a method used to calculate the number of days between two dates, such as the accrual of interest on a bond.
- Different day count conventions are used in the bond market, with the most common ones being the 30/360 convention, the Actual/360 convention, Actual/365 and the Actual/Actual convention.
- Each convention specifies how the number of days is counted and can affect the calculation of accrued interest.

DIRTY PRICE CALCULATION

Let's calculate the dirty price of a bond using an example:
Example:

Suppose you have a corporate bond with the following characteristics:

- Face Value (Par Value): $1,000
- Coupon Rate: 5% per annum (paid semiannually)
- Coupon Payment Dates: 30th June ; 31st Dec
- Bonds Settlement Date: August 15
- Day Count Convention: Actual/Actual
- Clean Priced of the Bond = 98.50 % of Par Value

DIRTY PRICE & CLEAN PRICE

To calculate the dirty price, you need to consider the clean price (quoted price) and the accrued interest.

1. Clean Price

Since the Bond is Trading at the clean price as quoted in the market at 98.50% of its face value. Therefore, the clean price is:

Clean Price = 0.9850 x $1,000 = $985.00

2. Accrued Interest

To calculate accrued interest, you need to determine the number of days that have elapsed since the last interest payment date (30th June 2025) up to the settlement date (15th August, 2025), using the Actual/Actual day count convention.
- 30th June'25 to 15th August'25: 46 days

Now, you can calculate the accrued interest using the formula:
Accrued Interest = (Accrued Days / Total Days in the Period) x (Annual Coupon Payment)

Accrued Interest = (46/365) x (0.05 x $1,000) = $ 6.30 (rounded to two decimal places)

DIRTY PRICE & CLEAN PRICE

To calculate the dirty price, you need to consider the clean price (quoted price) and the accrued interest.

3. Dirty Price

The dirty price is the sum of the clean price and the accrued interest.

Dirty Price = Clean Price + Accrued Interest

Dirty Price = $985.00 + $6.30 = $991.30 (rounded to two decimal places)

So, the dirty price of the bond on the settlement date, August 15, 2025, is $ 991.30. This is the total cost that the buyer would pay to acquire the bond, taking into account both the market price and the accrued interest earned by the seller up to that date.

CURRENT YIELD

In the context of a bond, the current yield is a financial ratio that measures the annual income return an investor can expect to receive from holding the bond at its current market price. It's a key metric for bond investors because it provides insights into the income-generating potential of the bond relative to its current trading price.

The formula for calculating the current yield of a bond is as follows:

Current Yield = {{Annual Interest or Coupon Payment}} / {Current Market Price of the Bond}} X 100

Where:

1. **Annual Interest or Coupon Payment:** This represents the annual interest payments made by the bond issuer to the bondholder. It's usually expressed as a fixed percentage of the bond's face value (par value). For example, if a bond has a face value of $1,000 and a coupon rate of 5%, the annual coupon payment is $50 ($1,000 * 0.05).

2. **Current Market Price of the Bond:** This is the current trading price of the bond in the secondary market. It may be different from the bond's face value (par value) which is quoted in % of Par Value and is determined by market supply and demand dynamics, prevailing interest rates, and other factors.

CURRENT YIELD

The current yield, expressed as a percentage, provides investors with an estimate of the bond's annual income return based on the bond's current market price. It's important to understand that the current yield does not take into account any potential capital gains or losses that may result from changes in the bond's market price over time.

Example of Calculating Current Yield for a Bond:

Let's say you have a corporate bond with the following characteristics:

- Face Value (Par Value): $1,000
- Coupon Rate: 5% per annum (paid semiannually)
- Current Market Price: 95.25% of Par Value = $952.50

CURRENT YIELD - CALCULATION

To calculate the current yield of this bond, you can use the formula:

Current Yield = {{Annual Coupon Payment}} / {Current Market Price}) X 100%

1. Calculate the Annual Coupon Payment:

Since the bond has a 5% coupon rate and is paid semiannually, the annual coupon payment is $1,000 (face value) * 0.05 (coupon rate) = $50 = $50

2. Use the Current Market Price: $952.50.

Now, calculate the current yield:

Current Yield = {Annual Coupon Payment}} / {Current Market Price}} X 100% ={\$50}} / {$952.50}}X100% = 5.25 %

So, the current yield of the bond is approximately 5.25%. This means that, based on the bond's current market price, an investor can expect to earn an annual return of about 5.25% in the form of coupon payments.

YIELD TO MATURITY (YTM)

Yield to Maturity (YTM)

- **Yield to Maturity (YTM)** is a crucial concept in bond investing. It represents the total return an investor can expect to receive from holding a bond until it matures, accounting for both its interest payments and any potential capital gains or losses if the bond is held to maturity.

- YTM is expressed as an annual percentage rate, and it's used by investors to assess the attractiveness of a bond investment.

YIELD TO MATURITY (YTM)

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Importance of Yield to Maturity (YTM):

1. Total Return Assessment: YTM provides a comprehensive measure of the potential return on a bond, considering both the income (interest payments) and the change in the bond's market price over its life.

2. Comparison Tool: YTM allows investors to compare the potential returns of different bonds with varying coupon rates, maturities, and market prices. It helps in evaluating which bonds offer better investment opportunities.

3. Investment Decision-Making: YTM is a critical factor in the decision-making process for bond investors. Investors typically seek bonds with YTMs that align with their investment objectives and risk tolerance.

4. Interest Rate Risk Assessment: YTM accounts for the sensitivity of a bond's market price to changes in interest rates. It helps investors assess the potential impact of interest rate fluctuations on the bond's value.

YIELD TO MATURITY (YTM) CALCULATION

The calculation of YTM can be complex because it involves finding the rate that makes the present value of a bond's future cash flows (coupon payments and face value) equal to its current market price.
The following formula represents the general approach to calculating Approximate YTM:

Yield to Maturity Approx $= \dfrac{C + \dfrac{F - P}{n}}{\dfrac{F + P}{2}}$

Where:
C = Annual Coupon / Interest Amount

Payment F = Face Value Or Redemption Value

P = Current Market Price

N – Years to Maturity

Bond Market Price above Par
Coupon rate higher than the market interest rate

Bond priced below Par Coupon rate lower than the interest rate

YIELD TO MATURITY (YTM) CALCULATION

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Interpretation of YTM:

If the **YTM** is **equal** to the bond's coupon rate, the bond is priced at par value, and the yield to maturity equals the coupon rate.

If the **YTM** is **greater than** the bond's coupon rate, the bond is priced at a discount (below par), indicating that investors can expect to earn a yield higher than the coupon rate by holding the bond to maturity.

If the **YTM** is **less than** the bond's coupon rate, the bond is priced at a premium (above par), suggesting that investors will earn a yield lower than the coupon rate if the bond is held to maturity.

Investors should consider YTM along with other factors, such as credit risk, reinvestment risk, and market conditions, when making bond investment decisions. YTM provides a valuable benchmark for evaluating bonds but should not be the sole determinant of investment choices.

DIFFERENCE BETWEEN CURRENT YIELD AND YIELD TO MATURITY

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Current Yield	Yield to Maturity
The current yield indicates the return on the bond based on its current market value.	The yield to maturity is an anticipated rate of the return associated with bonds till maturity
If a bond is bought at a discount of the face value, the YTM would be higher than that of the Current Yield as the discount raises the yield.	On the other hand, if a premium is paid for the bond, the YTM will be lesser than the current yield as the premium paid reduces the yield if the bond is redeemed at Par / Face Value

DIFFERENCE BETWEEN COUPON AND YIELD

Coupon	Yield
Coupon is the % Interest Rate return on a bond which is calculated upon Par or Face Value	Yield is the return % based on the price at which the bond was purchased.
In case of Fixed Coupon Rate Bonds, The Coupon Amount is fixed and never changes.	Yield changes based on the current market price of the bond.

INTEREST RATES

- Single most importance force in the bond market.
- There is an inverse relationship between Interest Rates & Bond Prices
- Changes in Interest rates affect the bond prices & they move in opposite directions.



Interest Rates ↑	Bond Prices ↓
Interest Rates ↓	Bond Prices ↑

- Bond markets are bullish when interest rates are low or falling.
- Bond markets are bearish when interest rates are high or rising.

BRIEFING OF YIELD, PRICE AND INTEREST RATE RELATION

Interest Rate	Price	Yield	Bearish Market
↑	↓	↑	

Interest Rate	Price	Yield	Bullish Market
↓	↑	↓	

NOMINAL INTEREST RATE VS REAL INTEREST RATE CONCEPT

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Nominal Interest Rate VS. Real Interest rate concept

Nominal interest rate do not consider or adjust the inflation rate, however Real Interest rate is inflation adjusted interest rate.

Example: If Interest rate is 5% and inflation rate is 2%, then nominal interest rate would be 5 % and real interest rate would be 3% (5% - 2%)





Securitization

SECURITIZATION

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Securitization is a financial process in which various types of illiquid assets like Mortgage Loans, Credit Care Loans, Student Loans or Receivables, are pooled together, converted into liquid tradable financial instruments called MBS/ABS (Mortgage-Backed Securities / Asset Backed Securities) and sold to investors. These securities represent a claim on the cash flows generated by the underlying assets. Securitization is commonly used in various industries, including Banking / Non-Banking / Real Estate / Infrastructure Power & Road Projects

Features of Securitization:

1. Pooling of Assets	2. Structuring Securities	3. Credit Enhancement
Securitization begins with the pooling of a large number of similar financial assets, such as mortgages, auto loans, credit card receivables, or even future cash flows from specific revenue streams like tolls from a highway.	The Orginator of Loans/Receivables like Banks often set up a special purpose vehicle or entity (SPV/SPE) and transfer structures the pooled assets to SPV which in turn issues financial securities to raise cash, typically in the form of MBS/ABS Bonds or Notes. These securities can be divided into different tranches, each with its risk and return profile.	To make the securities more attractive to investors, credit enhancement mechanisms may be used. This can include overcollateralization (adding more assets than needed to cover the securities), letters of credit, or insurance to protect against defaults on the underlying assets.

SECURITIZATION

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4. Rating Agencies

The securities are usually rated by credit rating agencies based on their creditworthiness and risk profile. Higher-rated tranches have lower risk and lower returns, while lower-rated tranches offer higher returns but come with higher risk.

5. Issuance and Sale

The issuer with the help of Investment Banks, sells these securities to investors in the capital markets. Investors may include banks, hedge funds, pension funds, and other institutional investors seeking fixed-income assets with varying levels of risk.

6. Cash Flows

The cash flows generated by the underlying assets, such as loan payments or interest income, are used to make payments to the investors. The structure of payments may differ among tranches, with senior tranches receiving payments before junior tranches.

7. Transfer of Risk

Securitization allows the issuer (often an originating institution like a Bank/NBFC) to transfer some or all of the risk associated with the underlying assets to the investors. This can help institutions manage their balance sheets and reduce exposure to specific types of loans.

EXAMPLE OF HOW SECURITIZATION WORKS

Let's consider a mortgage-backed securities (MBS) example:

1. Pooling Mortgages

A bank has a portfolio of mortgages it has originated. Instead of holding these mortgages on its books, the bank decides to securitize them. It pools together a large number of similar mortgages & Transfer these pools to Special Purpose Vehicle (SPV) which in turn issues MBS Securities to the Investors

2. Issuance and Sale

The SPV set up by bank sells the MBS Securities or Tranches (Collateral Mortgage Obligation) to various investors in the capital markets. Investors can choose tranches that align with their risk tolerance and return objectives.

3. Credit Enhancement

To make the MBS more attractive to investors, the bank may add credit enhancement, such as overcollateralization or a third-party guarantee, to some of the tranches.

4. Ratings

Credit rating agencies assess the risk of each tranche and assign credit ratings. The highest-rated tranches receive AAA ratings, indicating the lowest risk, while lower-rated tranches receive lower ratings.

EXAMPLE OF HOW SECURITIZATION WORKS

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5. Structuring MBS

The bank creates MBS securities backed by the cash flows from the pooled mortgages. These securities are structured into different tranches based on their risk characteristics. For instance, there may be Super Senior Tranche, Senior Tranche, Mezzanie Tranche & Equity Tranche with different credit quality and returns.

6. Cash Flows

As homeowners make mortgage payments, the cash flows are passed through the MBS structure. Senior tranches receive payments first, followed by subordinated tranches. If there are defaults on the mortgages, the impact is absorbed by the lowest-rated (most junior) tranches first.

In this example, the bank has effectively transferred the credit risk associated with the mortgages to investors in the MBS market, allowing the bank to free up capital for additional lending while providing investors with the opportunity to invest in a diversified portfolio of mortgage-related assets.



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THANK YOU !!!

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